

Entrepreneurship: Commercializing Our Pulmonary Delivery Platform

This section elaborates on the entrepreneurial aspects of our project, focusing specifically on the pulmonary delivery system. We analyze the market opportunity, competitive advantages, intellectual property strategy, and commercialization pathway, all grounded in the chemistry and engineering of our nebulization technology.

First is about our “Targeted Pulmonary Delivery via Aerosolization”. Our delivery system employs a vibrating mesh nebulizer that precisely controls aerosol particle size (1–5 μm) to ensure engineered bacteria deposit in the bronchial and distal airways where SCLC commonly resides. Bacterial viability during nebulization is maintained >85% through an optimized protectant formulation (trehalose, mannitol, vitamin C). Furthermore, surface expression of P30 adhesin and DLL3 nanobodies enables bronchial epithelial anchoring and active tumor cell recognition, creating a multi-layered delivery strategy of “physical deposition + biological targeting.” This system not only enhances local drug concentration but also significantly reduces systemic exposure.

Secondly, we have Nebulized inhalation that is non-invasive, home-based, and takes only 10–15 minutes per session, eliminating frequent hospital visits. Compared to intravenous infusion (3–5 hours per visit) and subcutaneous injection (painful, requires training), it significantly improves quality of life. Also, the protectant formulation (synergistic effects), lyophilization process, and particle size control methods are patent-pending, creating multi-layer IP protection covering compositions, methods, and uses. Competitors cannot easily circumvent. Lastly, the platform technology can be adapted to other respiratory diseases (e.g., pneumonia, pulmonary fibrosis) by simply swapping targeting modules,

reducing follow-up R&D costs.

More importantly, We have built a patent portfolio around the chemical innovations of our delivery system:

For Protectant composition, We made a specific ratio of trehalose, mannitol, and vitamin C exhibits synergistic protection under nebulization stress, resulting in significantly higher post-nebulization viability than single components (invention patent filed).

For Lyoprotectant formulation, we choose "Trehalose + polyvinylpyrrolidone"(PVP) form and successfully accelerated its stability.